ECEN-629: Homework 3

Due on October 21, 2025 at 11:59 pm

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Problem 1

Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is convex, and define

$$g(x,t) = f(x/t), \quad \mathbf{dom}\, g = \{(x,t) \mid x/t \in \mathbf{dom}\, f,\ t > 0\}.$$

Show that g is quasiconvex, and then write down the optimization procedure similarly to that on page 17 of the slides of Chapter 7, specialized to this problem.

Hint: consider the function $t \cdot f(x/t)$.

Solution

Problem 2

Some non-convex problems can also bea easy to solve. Find the optimal solution for the following problems in closed form.

(a) For $x \in \mathbb{R}^n$

$$\begin{aligned} \text{minimize:} \quad & \|x\|_2 \\ \text{subject to:} \quad & 10 \geq \|x\|_1 \geq 1. \end{aligned}$$

(b) Let $x = (x_1, x_2) \in \mathbb{R}^2$

$$\begin{aligned} \text{minimize:} \quad & x_1 + x_2 \\ \text{subject to:} \quad & x_1 + 3x_2 = 5; \\ & (x_1 - 3)^2 + (x_2 - 4)^2 \geq 25. \end{aligned}$$

Solution

Problem 3

Huber's penalty function is defined as

$$\phi(u) = \begin{cases} u^2, & |u| \leq M, \\ M(2|u| - M), & |u| > M. \end{cases}$$

(a) Is the function $\phi(u)$ convex in general?

(b) Show that the minimization problem

$$\text{minimize:} \quad \sum_{i=1}^{n} \phi(a_i^T x - b_i)$$

is equivalent to the following problem

$$\begin{aligned} \text{minimize:} \quad & \sum_{i=1}^{n} (u_i^2 + 2Mv_i) \\ \text{subject to:} \quad & -u - v \preceq Ax - b \preceq u + v \\ & 0 \preceq u \preceq M\mathbf{1} \\ & v \succeq 0. \end{aligned}$$

(c) A general form of Huber's penalty function with parameters (α, β, γ) is

$$\phi(u) = \begin{cases} \alpha \cdot u^2, & |u| \leq M, \\ \beta M(\gamma|u| - M), & |u| > M. \end{cases}$$

For any fixed value of $\alpha > 0$, find the valid assignment of (β, γ) in terms of α such that the penalty function is convex.

Solution

Problem 4

Write the following problems as equivalent LPs:

(a)

$$\begin{aligned} \text{minimize:} \quad & \|x\|_\infty \\ \text{subject to:} \quad & \|Ax - b\|_1 \leq 1 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

(b)

$$\begin{aligned} \text{minimize:} \quad & \|x\|_\infty \\ \text{subject to:} \quad & \|Ax - b\|_\infty \leq 1 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

(c)

$$\begin{aligned} \text{minimize:} \quad & \|x\|_1 \\ \text{subject to:} \quad & \|Ax - b\|_\infty \leq 1 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

(d)

$$\begin{aligned} \text{minimize:} \quad & \|x\|_1 \\ \text{subject to:} \quad & \|Ax - b\|_2 \leq 1 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

For the problems above, what if we exchange the objective function and the constraint, can you still write down the problems as LPs?

Solution

Problem 5

Consider the optimization problem in $\mathbb{R}^2$

$$\begin{aligned} \text{minimize:} \quad & f_0(x_1, x_2) \\ \text{subject to:} \quad & (x_1 - 1)^2 + (x_2 - 2)^2 \leq 2 \\ & 2x_1 + x_2 \geq 3 \\ & x_1 \geq 0,\ x_2 \geq 0. \end{aligned}$$

First make a sketch of the feasible set. For each of the following objective functions, give the optimal set and the optimal value, using two methods: geometrically and the first-order optimality condition. Note that you should not use KKT directly here, but the first-order condition in Chapter 4.

(a) $f_0(x_1, x_2) = x_1 + x_2$;

(b) $f_0(x_1, x_2) = x_1^2 + x_2^2$.

Solution

Problem 6

Consider the optimization problem

$$\begin{aligned} \text{minimize:} \quad & f_0(x_1, x_2) \\ \text{subject to:} \quad & x_1 + x_2 \geq 2 \\ & 2x_1 + x_2 \geq 3 \\ & x_1 \geq 0, \ x_2 \geq 0. \end{aligned}$$

First make a sketch of the feasible set. For each of the following objective functions, give the optimal set and the optimal value, using two methods: geometrically and the first-order optimality condition.

(a) $f_0(x_1, x_2) = 3x_1 + 2x_2$;

(b) $f_0(x_1, x_2) = \max\{x_1, x_2\}$.

Solution

Problem 7

Give explicit solutions for the following LP problems over $\mathbb{R}^n$

(a)

$$\begin{aligned} \text{minimize:} \quad & c^T x \\ \text{subject to:} \quad & a^T x \leq b, \end{aligned}$$

where $a \neq 0$.

(b)

$$\begin{aligned} \text{minimize:} \quad & c^T x \\ \text{subject to:} \quad & \mathbf{1}^T x \leq \alpha \\ & 0 \cdot \mathbf{1} \preceq l \preceq x \preceq u. \end{aligned}$$

Solution

Problem 8

Find the dual optimization problems for the following optimization problems

(a)

$$\begin{aligned} \text{minimize:} \quad & c^T x \\ \text{subject to:} \quad & Ax = b \\ & l \preceq x \preceq u, \end{aligned}$$

where $x \in \mathbb{R}^n$ is the variable, and $l, u \in \mathbb{R}^n$

(b) Recall last time you wrote the following problem as an equivalent LP:

$$\begin{aligned} \text{minimize:} \quad & \|x\|_\infty \\ \text{subject to:} \quad & \|Ax - b\|_\infty \leq 1. \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Find the dual problem for the equivalent LP.

(c)

$$\begin{aligned} \text{minimize:} \quad & \sum_{i=1}^{n} x_i \log x_i \\ \text{subject to:} \quad & Ax \succeq b, \\ & c^T x = d, \end{aligned}$$

whose domain is $\mathbb{R}^n_+$, i.e., $x_i > 0$ for all $i = 1, 2, \ldots, n$. Assume $c \succeq 0$.

(d)

$$\begin{aligned} \text{minimize:} \quad & x^T Q x \\ \text{subject to:} \quad & Ax = b, \end{aligned}$$

where Q is a positive definite matrix.

Solution